

3.0 Milestone Assessment: Network Monitoring, SNMP, Ports, Protocols, and Services

Introduction

Modern networks depend on continuous communication between devices, applications, and services to support business operations and user connectivity. To maintain reliability, security, and performance, network administrators must understand how networks normally behave, how communication occurs across ports and protocols, and how monitoring technologies help identify problems before they impact users.

Networking operates as an interconnected system where physical infrastructure, communication protocols, and monitoring technologies all work together to support efficient data transmission. Weaknesses in one area, like poor physical installation, excessive traffic, or failed monitoring, can negatively affect the entire network.

To fully understand how modern networks function, it is important to examine how network baselining and Simple Network Management Protocol, SNMP, support monitoring and troubleshooting, while ports, protocols, services, and traffic types enable communication between devices and applications.

Baselining and Abnormal Behavior

Network baselining is the process of measuring and documenting normal network performance over time. A baseline creates a reference point that allows administrators to understand what healthy network activity typically looks like during daily operations.

Administrators commonly monitor bandwidth usage, latency, packet loss, central processing unit (CPU) and memory utilization, interface statistics, and traffic patterns. By observing this information over time, network administrators can identify trends and establish normal operating conditions.

The importance of baselining is that it allows administrators to quickly recognize abnormal behavior. Without a baseline, it is difficult to determine whether network activity is expected or potentially harmful.

For example, if a network normally operates at 35% bandwidth utilization during business hours but suddenly reaches 90% utilization late at night, administrators can

investigate possible causes such as malware, unauthorized downloads, denial-of-service attacks, failed backups, or network misconfigurations. This may have been a missing component in the latest Instructure security incident.

Other abnormal behavior may include:

- Excessive broadcast traffic
- Increased packet loss
- High CPU utilization
- Unusual DNS traffic
- Interface failures
- Unauthorized login attempts
- Unexpected spikes in latency

Baselining also supports troubleshooting, performance optimization, capacity planning, and security analysis. By comparing current activity to normal operating conditions, administrators can identify problems more quickly and reduce downtime.

Simple Network Management Protocol (SNMP)

Simple Network Management Protocol (SNMP) is used to monitor and manage network devices such as routers, switches, servers, firewalls, printers, and wireless access points. SNMP operates using three primary components:

- **SNMP Manager:** acts as the central monitoring system that collects and analyzes information from network devices.
- **SNMP Agent:** software installed on managed devices that gathers local device information and communicates with the manager.
- **Management Information Base (MIB):** a structured database containing operational statistics and device information.

SNMP primarily functions through polling and traps, which occur when the SNMP manager regularly requests information from network devices, including: interface status, bandwidth utilization, CPU usage, memory usage, error counts and device temperature.

Traps are automatic alerts sent by devices when important events occur, such as device failures, interface outages, authentication failures, power supply problems or high CPU utilization.

SNMP plays an important role in network monitoring because it provides centralized visibility into network performance and device health. Administrators can identify bottlenecks, detect outages, monitor long-term trends, and respond to developing problems before users experience major disruptions.

SNMPv3 is the preferred version in modern networks because it includes authentication and encryption features that improve security during remote monitoring.

Physical Installation Factors and Network Reliability

Physical installation directly impacts network performance, reliability, and security. Even well-configured networks can experience poor performance if the physical infrastructure is improperly installed or maintained. Several important factors affect physical network installations.

1. Cable quality and cable type determine bandwidth capabilities and resistance to interference. Higher-grade cabling, such as Cat6 or fiber optic cabling, supports higher speeds and improved reliability.
2. Distance limitations must also be considered. Ethernet cables typically support a maximum distance of 100 meters before signal degradation begins to occur.
3. Environmental conditions such as heat, humidity, and electromagnetic interference (EMI) can negatively affect network performance. Proper ventilation, shielding, and equipment placement help reduce these risks.
4. Proper cable management and labeling simplify troubleshooting and maintenance while improving airflow and reducing accidental disconnections.
5. Power protection is also critical. Uninterruptible Power Supplies (UPS) and surge protectors help protect equipment during outages and electrical fluctuations.
6. Physical security is equally important because unauthorized access to networking equipment can compromise the entire network infrastructure. Equipment should be secured using locked rooms, restricted access, and monitoring systems.

A well-designed physical installation supports stable communication, minimizes downtime, and improves long-term scalability.

Ports, Protocols, Services, and Traffic Types

Networks rely on ports, protocols, and services to allow devices and applications to communicate effectively.

1. Protocols define the rules for communication between systems.
2. Ports identify specific services or applications running on a device.
3. Services perform functions such as web communication, email delivery, file transfers, and remote management.

These technologies work together to support modern network functionality.

Traffic Types and Network Communication

Most modern network traffic uses either TCP or UDP. Transmission Control Protocol (TCP) is connection-oriented and focuses on reliable communication. TCP uses acknowledgments, sequencing, and error checking to ensure data is delivered correctly. Services such as HTTPS, email, file transfers, and SSH commonly use TCP because reliability is critical.

User Datagram Protocol (UDP) is connectionless and focuses on speed and efficiency. UDP does not guarantee delivery, making it useful for applications where low latency is more important than perfect reliability. Streaming media, VoIP, gaming, and DNS commonly use UDP. These technologies interact continuously throughout the communication process.

For example, when a user visits a secure website:

- DNS uses port 53 to resolve the website name into an IP address
- TCP establishes a reliable connection
- HTTPS uses port 443 to encrypt communication
- The web service exchanges data securely between the client and server

When administrators remotely manage a router:

- TCP provides reliable transport
- Port 22 identifies the SSH service
- SSH encrypts the remote management session

Together, ports, protocols, and services create the foundation that allows devices, applications, and networks to communicate securely and efficiently.

Conclusion

Network baselining and SNMP are essential technologies for maintaining reliable and secure networks. Baselining establishes normal operating conditions that help administrators identify abnormal behavior, while SNMP provides centralized monitoring and alerting for network devices.

Physical installation factors such as cable management, environmental controls, power protection, and physical security directly impact network stability and performance.

Ports, protocols, services, and traffic types form the foundation of network communication by allowing systems and applications to exchange data efficiently. Understanding how these technologies interact allows administrators to troubleshoot problems, optimize performance, improve security, and maintain reliable communication across modern enterprise networks.

Common Networking Ports and Protocols

Port	Protocol	Service	Purpose
20/21	FTP	File Transfer Protocol	Transfers files between systems
22	SSH	Secure Shell	Secure remote administration
23	Telnet	Remote terminal access	Insecure remote management
25	SMTP	Simple Mail Transfer Protocol	Sends email
53	DNS	Domain Name System	Resolves domain names
67/68	DHCP	Dynamic Host Configuration Protocol	Assigns IP addresses automatically
69	TFTP	Trivial File Transfer Protocol	Lightweight file transfers
80	HTTP	Hypertext Transfer Protocol	Web communication
110	POP3	Post Office Protocol v3	Retrieves email
123	NTP	Network Time Protocol	Synchronizes system clocks
143	IMAP	Internet Message Access Protocol	Accesses stored email
161/162	SNMP	Simple Network Management Protocol	Network monitoring and alerts
389	LDAP	Lightweight Directory Access Protocol	Directory services
443	HTTPS	Secure HTTP	Encrypted web communication
445	SMB	Server Message Block	File and printer sharing
514	Syslog	Logging service	Centralized logging
636	LDAPS	Secure LDAP	Encrypted directory services
3389	RDP	Remote Desktop Protocol	Remote graphical administration

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